

# ABHINAV KAUSHIK

Senior AI/ML Engineer

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## SUMMARY

Senior AI/ML Engineer with 8+ years of experience delivering production-grade machine learning solutions across healthcare, telecom, and enterprise domains. Proven expertise in building and scaling recommendation, semantic search, and personalization systems using embeddings, LLMs, RAG, LangChain and LangGraph. Strong background in MLOps practices—including MLflow, Docker, Kubernetes, Airflow and GCP—ensuring reliable model deployment and continuous experimentation. Adept at collaborating with product, data engineering and analytics teams to translate business needs into impactful AI products.

## TECHNICAL SKILLS

**AI / GenAI:** GenAI, LLM, RAG, Agentic AI, LangChain, LangGraph, Claude, OpenAI, Embeddings, Tool Orchestration, Cross-Encoder, Transformers, NLP, Generative AI

**Machine Learning:** XGBoost, LightGBM, Random Forest, LSTM, SBERT, DBSCAN, Time Series Forecasting, Demand Forecasting, Anomaly Detection, Statistical Modeling, Regression, Deep Learning

**MLOps / Cloud:** MLflow, Docker, Kubernetes, Jenkins, Bamboo, Bitbucket, DVC, Terraform, Airflow, GCP, Azure

**Search / Observability:** Elasticsearch, Kibana, Logging, Log Analysis, Apache Solr, Lucene Indexing, Search and Information Retrieval

**Programming / Data:** Python, SQL, PySpark, Pandas, NumPy, Flask, Hive, SparkQL, REST API, Pydantic, Unix

**Data Platforms:** Databricks, Apache Spark, Airflow

**Databases:** TimescaleDB, MySQL, Vector DB, pgvector, ArangoDB, Cosmos DB

**BI / Testing:** Power BI, QlikSense, SAP HANA, Selenium

## PROFESSIONAL EXPERIENCE

**Senior AI Engineer** | Optum.AI

Bengaluru, India | Nov 2025 – Present

- Enhanced dense retrieval performance by training and evaluating a cross-encoder with positive and negative examples for AARP survey data, achieving a 20% increase in retrieval accuracy.
- Engineered a LangGraph-based analytics system for AARP medical plans, combining stateful multi-agent reasoning with real-time chart generation and reducing manual analysis effort by 98%.
- Built and integrated a lab extraction agent with an existing Laboratory Information Management System (LIMS) using Claude Opus 4.7 and Amazon Textract, automating work associated with 800 Jira tickets and PRD tasks.

**Senior Machine Learning Engineer** | Viavi Solutions

Chennai, India | Sep 2024 – Nov 2025

- Facilitated development of an AI-powered Project Discovery Assistant that analyzes client transcripts, summaries, and uploaded documents using LangChain and LangGraph, achieving a 90% match rate of technical use cases with existing organizational solutions.
- Built 30+ AI application prototypes and mockups for clients across AIOps and GeoPerformance, translating client requirements into functional POCs for solution validation and demonstrations.
- Deployed a production-grade SLA breach prediction system using Random Forest and LightGBM with FastAPI and MLflow, integrating GPT-4-powered reasoning for real-time root-cause analysis and remediation guidance, resulting in a 67% reduction in downtime.

**Data Scientist** | Viavi Solutions

Bengaluru, India | Nov 2021 – Sep 2024

- Applied Sentence-BERT and DBSCAN to enhance semantic understanding and cluster alarm messages across graph-based AIOps infrastructure, achieving a 73.5% improvement in semantic alarm classification.
- Applied Isolation Forest and Holt-Winters for anomaly detection and demand forecasting across 500+ telecommunications network devices using 15+ KPIs, supporting earlier identification of abnormal time-series behavior and proactive network planning.
- Delivered technical POCs and consulting sessions on data services and products for 15+ clients across 3+ AIOps and ML platforms, contributing to an 11% increase in POC and mentoring revenue.

**Data Analyst** | United Health Group

Gurugram, India | Jun 2021 – Nov 2021

- Collaborated with stakeholders on ad-hoc analysis and customer-retention dashboards using Hive, SQL, and Power BI.
- Developed a Python- and Flask-based parser to automate data extraction, parsing, and transformation into structured data, supporting downstream analytics and reporting.

**Data Engineer** | Nippon Data Systems

Gurugram, India | Nov 2019 – May 2021

- Engineered SQL and PySpark-based data pipelines to automate BAU medical-audit reporting, transforming and aggregating large healthcare datasets and reducing manual reporting effort by 95%.
- Built real-time observability using Elasticsearch for centralized log ingestion and analysis, with Kibana dashboards for monitoring system activity and user events, improving traceability efficiency by 80% for the COVID QRT team.

- Developed a Flask-based enterprise search engine using Apache Solr and Lucene indexing, enabling centralized search and retrieval across organizational data for enterprise-wide use cases.
- Associate Consultant** | SaksoftGurugram, India | Dec 2017 – Nov 2019
- Conceptualized customer dashboards using QlikSense, SQL, and SAP HANA for financial and product insights, resulting in a 40% increase in user engagement.
  - Developed an XGBoost-based propensity model to predict the probability of customers initiating a personal loan application within 7 days of contact, enabling targeted outreach and customer prioritization.
  - Leveraged Selenium and Python to automate performance testing and evaluate application response times, resulting in a 90% reduction in testing effort.

**PROJECTS**

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**Think Tree - ML Study Platform** – Live Platform | <https://think-tree-ai.vercel.app/>

- Built a non-profit AI-powered ML study platform for people with ADHD and context-switching challenges, designed to help users organize learning and maintain study context.

**Financial Fraud Detection** – Mar 2025 | GitHub

- Developed an AI-driven fraud detection and reasoning workflow using the JPMorgan Synthetic Dataset, combining LightGBM-based modeling with LLM explanations to support interpretable financial-risk analysis.

**E-Commerce Personalization & Search Prototype** – Portfolio Project

- Developed a prototype recommendation and re-ranking pipeline for an online beauty retailer using product embeddings and a two-stage retrieval-ranking architecture.
- Integrated LangChain for natural language query understanding and RAG to generate dynamic product descriptions, showcasing LLM-powered shopping assistance capabilities.

**RFP Multi-Agent Proposal Automation Platform** – GitHub

- Built a multi-agent RFP automation platform using an LLM-driven supervisor to orchestrate intent detection, retrieval, proposal generation, document matching, and market research with RAG over historical proposals.

**EDUCATION**

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|--------------------------------------------------------------------------------------|---------------------|
| <b>Master of Science, Machine Learning and AI</b>   Liverpool John Moores University | Jan 2020 – Jan 2022 |
| <b>Bachelor of Technology, Information Technology</b>   Jamia Hamdard                | Aug 2013 – Aug 2017 |

**ADDITIONAL INFORMATION**

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- German: A1 (in progress)
- Network Intern, Airport Authority of India
- Hadoop Intern, Ducat Technologies